Generate Equivalent Expressions

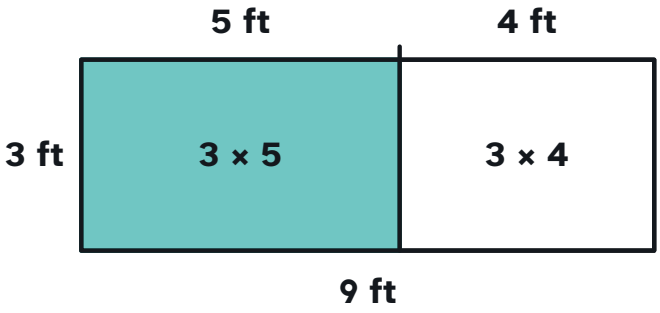
Expand to remove parentheses, or factor by pulling out the greatest common factor.

Name _____ **Date** _____

Tonight's spark

A rug is 3 feet tall and made of two strips. One strip covers 15 square feet. The other covers 12 square feet. How wide is the whole rug?

Guess first: is the rug wider than 5 feet, or narrower?



Our class model: pull out the GCF

Try it

Work it out on paper. Write one answer on each line.

1 **Expand 3(x + 2).**

A. 3x + 2 **B.** 3x + 5 **C.** 3x + 6 **D.** x + 6

My answer _____

2 **Which expression equals 5(2 + 7)?**

A. 5 × 2 + 5 × 7 **B.** 5 × 2 + 7 **C.** 2 + 5 × 7 **D.** 5 + 2 + 7

My answer _____

3 **What is the greatest common factor (GCF) of 12 and 18?**

A. 3 **B.** 6 **C.** 2 **D.** 36

My answer _____

4 **Expand 2(4x + 3).**

A. 8x + 3 **B.** 4x + 6 **C.** 8x + 5 **D.** 8x + 6

My answer _____

5 **Factor 8x + 12 using the greatest common factor.**

A. 4(2x + 3) **B.** 2(4x + 6) **C.** 4(2x + 12) **D.** 12(x + 1)

My answer _____

6 **A rug is 3 feet tall. Its width is split into a part x feet wide and a part 4 feet wide. Which expression shows the total area as the sum of the two parts?**

A. 3x + 4 **B.** x + 12 **C.** 3x + 12 **D.** 3 + x + 4

My answer _____

Talk about it

Did the outside number reach every term inside the parentheses? What is the greatest common factor (GCF) of the two terms?

Home mission

Set out 4 plates. Put 3 crackers and 2 grapes on each. Can your student say the total two ways, 4(3 + 2) and 4 × 3 + 4 × 2? Then push the food together: 12 crackers and 8 grapes. What number is shared by both?

If we get stuck, ask

- Did the outside number reach every term inside?
- What is the biggest number that divides both terms?
- Can you check by doing the opposite move?

Do it on a screen

Scan or type the link. Same lesson, with hints.

eduwonderlab.com/lessons/6-8-part2/homework.html

How did it feel? (optional)

Tricky Getting there Got it